Difference of two squares

Task A: Multiplying binomials

1) Fill in the area models for these products of two binomials. Write your answer as a simplified expression

a)

	x	-1
x	x^2	$-x$
1	x	-1

$(x + 1)(x - 1)$

$\equiv x^2 - x + x - 1$

$\equiv x^2 - 1$

c)

	x	-2
x	x^2	$-2x$
-2	$-2x$	4

$(x - 2)(x - 2)$

$\equiv x^2 - 2x - 2x - 4$

$\equiv x^2 - 4x - 4$

b)

	x	10
x	x^2	$10x$
-10	$-10x$	-100

$(x - 10)(x + 10)$

$\equiv x^2 - 10x + 10x - 100$

$\equiv x^2 - 100$

d)

	x	-0.5
x	x^2	$-0.5x$
0.5	$0.5x$	-0.25

$(x + 0.5)(x - 0.5)$

$\equiv x^2 - 0.5x + 0.5x - 0.25$

$\equiv x^2 - 0.25$

2) Which of these products of two binomials can be simplified to an expression with exactly two terms?

a) $(x + 9)(x - 8)$

b) $(x - 9)(x - 9)$

c) $(x - 6)(x + 5)$

d) $\boxed{(x - 13)(x + 13)}$

e) $(x + 7)(x + 7)$

f) $\boxed{(x + 11)(x - 11)}$

g) $\boxed{(x - 2.4)(x + 2.4)}$

h) $(x - 5)(x + 5.5)$

i) $\boxed{(x + \frac{11}{2})(x - 5.5)}$

j) $\boxed{(x - 0.8)(x + \frac{4}{5})}$

k) $(x + \frac{3}{8})(x - 0.37)$

l) $(x - \frac{17}{20})(x - 1.85)$

3) Match a binomial in the left column with a binomial in the middle column and find its product in the end column.

a)	$(x - 6)$	p)	$(x + 6)$	x)	$x^2 - 36$
b)	$(x - 5)$	k)	$(x - 5)$	u)	$x^2 - 10x + 25$
c)	$(x - 2)$	o)	$(x + 1)$	v)	$x^2 - x - 2$
d)	$(x + 1)$	n)	$(x - 1)$	t)	$x^2 - 1$
e)	$(x + 2)$	m)	$(x - 2)$	s)	$x^2 - 4$
f)	$(x + 4)$	i)	$(x - 9)$	w)	$x^2 - 5x - 36$
g)	$(x + 7)$	j)	$(x - 7)$	q)	$x^2 - 49$
h)	$(x + 9)$	l)	$(x - 4)$	r)	$x^2 + 5x - 36$

Task B: Identifying difference of two squares

1) Which of these products of two binomials can be written as the difference of two squares?

a) $(x + 1)(x - 4)$

b) $(y + 1)(x - 1)$

c) $\boxed{(a - 46)(a + 46)}$

e) $\boxed{(x - y)(x + y)}$

f) $(3a - b)(3a - b)$

g) $\boxed{(10 + x)(10 - x)}$

$\equiv 10^2 - x^2 \equiv 100 - x^2$

i) $\boxed{(xy - 3)(xy + 3)}$

j) $(3x - 2y)(-3x + 2y)$

One of the terms in each bracket needs to be the same and the other two should be a zero pair. Both of these are zero pairs.

2) Fill in the area models for these products of two binomials. Write your answer as the difference of two squares.

a)

y^2	$20y$
$-20y$	-400

$(y + 20)(y - 20)$

$\equiv y^2 - 20^2$

$\equiv y^2 - 400$

c)

x^2	$2xy$
$-2xy$	$-4y^2$

$(x - 2y)(x + 2y)$

$\equiv x^2 - (2y)^2$

$\equiv x^2 - 4y^2$

b)

x^2	$10x$
$-10x$	-100

$(x - 10)(x + 10)$

$\equiv x^2 - 10^2$

$\equiv x^2 - 100$

d)

a^2	$3ab$
$-3ab$	$-9b^2$

$(a - 3b)(a + 3b)$

$\equiv a^2 - (3b)^2$

$\equiv a^2 - 9b^2$

e)

9	$-3x$
$3x$	$-x^2$

$(3 + x)(3 - x)$

$\equiv 3^2 - x^2$

$\equiv 9 - x^2$

g)

x^2	$4x$
$-4x$	-16

$(x - 4)(x + 4)$

$\equiv x^2 - 4^2$

$\equiv x^2 - 16$

f)

$4y^2$	$6xy$
$-6xy$	$-9x^2$

$(2y - 3x)(2y + 3x)$

$\equiv (2y)^2 - (3x)^2$

$\equiv 4y^2 - 9x^2$

h)

$25a^2$	$15ab$
$-15ab$	$-9b^2$

$(5a + 3b)(5a - 3b)$

$\equiv (5a)^2 - (3b)^2$

$\equiv 25a^2 - 9b^2$